

No.	Questions	Marks	CO	BL	PI
Q.1	NOTE: Attempt all questions. All questions carry equal Marks. Reduce the matrix to normal form and hence find the rank of the matrix $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$	5	CO 1	2	1.1.1
Q.2	Investigate the values of λ and μ so that the equations $2x+3y+5z=9$, $7x+3y-2z=8$, $2x+3y+\lambda z=\mu$ have (i) no solution, (ii) a unique solution, (iii) an infinite number of solutions. OR (a) Find the inverse of a matrix by Gauss Jordan Method where $A = \begin{bmatrix} 4 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. (b) Is the system of vectors linearly dependent $X_1 = (1, 2, -2)$, $X_2 = (-1, 3, 0)$, $X_3 = (0, -2, 1)$.	5	CO1	2	1.1.1
Q.3	Solve $(x-y)dy + y^2dx = 0$.	5	CO 2	2	1.1.1
Q.4	Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$. Solve $(1+4xy+2y^2)dx + (1+4xy+2x^2)dy = 0$.	5	CO 2	2	1.1.1

Shri G. S. Institute of Technology and Science, Indore
Department of Applied Mathematics and Computational Science

B. Tech Ist Year (April - June 2022) Section (F - J)
MA 10001 (MATHEMATICS-I), Mid Term Test-II

TIME: 01 Hour

NOTE: Attempt all questions. All questions carry equal Marks

[Max Marks: 20]

No.	Questions	Marks	CO	BL	PI
Q.1	A wire of length b is cut into two parts which are bent in the form of a square and circle respectively. Find the least value of the sum of the areas so found. (using Lagrange's method). 12.14150265	4	CO2	3	1.1.1
Q.2	Examine the function $f(x, y) = y^2 + 4xy + 3x^2 + x^3$ for extreme values. $00 \quad 2, -\frac{4}{3}$	4	CO2	3	1.1.1
Q.3	Trace the curve $y^2 x^2 = x^2 - a^2$.	4	CO4	2	1.1.1
Q.4	Find the whole area of the curve $r = a \sin 3\theta$.	4	CO4	2	1.1.1
Q.5	Let $[B, +, \cdot, ']$ be a Boolean algebra and $a, b \in B$, then show that $(a \cdot b)' = a' + b'$	4	CO5	2	1.1.1

July-Dec. 2019-I

Shri G.S. Institute of Technology & Science, Indore
Department of Applied Mathematics & Computational Science
B.TECH. I year, I-Mid Term Test (July-Nov 2019)(Sec F- J)
MA-10501: Mathematics-II

Time: I Hour

Max. Marks: 20

Note: Attempt any four questions. All questions carry equal marks.

Q.1 If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then find two non-singular matrices P and Q such that PAQ is in normal form. Hence find rank of A. (05)

Q.2 For what values of λ and μ the system of linear equations:
 $x + y + z = 6$, $x + 2y + 5z = 10$, $2x + 3y + \lambda z = \mu$
has (i) a unique solution (ii) no solution (iii) infinite solutions. Also find the solution for $\lambda = 2$ and $\mu = 8$. (05)

Q.3 Solve differential equation $x \frac{dy}{dx} + y = y^2 \log x$. (05)

Q.4 Solve $\frac{d^2y}{dx^2} + 4y = x^2 + \cos 2x$. (05)

Q.5 Solve $(xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0$. (05)

Department of Applied Mathematics & Computational Science

B.E. I Year (Section A - E)

Mid-Term Test-I (Dec. 2018-April 2019)

MA-10501 : Mathematics- II

Time: 01 Hour

Max. Marks: 20

Note: Attempt any four questions. All questions carry equal marks.

Q.1 By using Gauss-Jordan method, find the inverse of the matrix.

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & 4 \end{bmatrix}$$

(05)

Q.2 Let $A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$. Find matrices P and Q is in the normal form. Also find rank of A. (05)

Q.3 Solve the following: $(xy \sin xy + \cos xy)ydx + (xy \sin xy - \cos xy)x dy = 0$ (05)

Q.4 Solve the following: $\frac{dy}{dx} = \frac{y+x-2}{y-x-4}$. (05)

Q.5 Solve the following: $\sqrt{1-y^2} dx = (\sin^{-1} y - x) dy$ (05)

Department of Applied Mathematics & Computational Science
B.E.I year, I-Mid Term Test (Oct 2018)
MA-10501:-Mathematics-II (Section F-J)

Time:1 Hour

Max.Marks:20

Note: Attempt any four questions. All questions carry equal marks.

- Q.1 (i) Solve $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$ (2.5+2.5)
(ii) Prove that a square real matrix A can be written as sum of a symmetric matrix and a skew-symmetric matrix.

Q.2 Find the rank of the matrix $A = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 2 & -1 & 3 & 2 \\ 3 & -5 & 2 & 2 \\ 6 & -3 & 8 & 6 \end{bmatrix}$ by reducing it to normal form. (05)

Q.3 Solve $y'' - 2y' + 3y = \cos x + x^2$. (05)

Q.4 Solve $x \frac{dy}{dx} + y \log y = xye^x$. (05)

Q.5 For what values of k, the equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ has a solution and find them. (05)

Department of Applied Mathematics & Computational Science
B.E.I year, II-Mid Term Test (Nov. 2018)
MA-10501:-Mathematics-II (Section F-J)

Time:1 Hour

Max.Marks:20

Note: Attempt any four questions. All questions carry equal marks.

- Q.1 The probability that a bomb dropped from a plane will strike the target is $1/5$. If six bombs are dropped, find the probability that : (05)

(i) Exactly two will strike the target.

(ii) At least two will strike the target.

Q.2 Fit a second degree parabola $y = a + bx + cx^2$ to the following data: (05)
(x, y): (-1, 2), (0,0), (0, 1), (1, 2)

- Q.3 An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident? (05)

Department of Applied Mathematics & Computational Science
 Mid-Term Test-I (July 2016-Dec.2016)
 B.E. I Year (Section F to J)
 MA 10501 : Mathematics- II

Time: 01 Hour

Note: Attempt any three questions. All questions carry equal marks.

Max. Marks: 15

Q.1	Solve $(xy^2 - e^{1/x^3})dx - x^2y dy = 0$	(05)
Q.2	Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4\cos^2x + e^{2x}$	(05)
Q.3	Find the rank of the matrix: $\begin{bmatrix} 2 & 3 & -2 & 4 \\ 3 & -2 & 1 & 2 \\ 3 & 2 & 3 & 4 \\ -2 & 4 & 0 & 5 \end{bmatrix}$	(05)
Q.4	For what values of k the system of equations: $\begin{aligned} x + y + z &= 1 \\ 2x + y + 4z &= k \\ 4x + y + 10z &= k^2 \end{aligned}$ has a solution.	(05)

xy^4
 $2xy$
 $2(x^2y + x + y^2)$
 $2xy$ $4xy^2$ y
 $2y^4 + y^2$

Department of Applied Mathematics & Computational Science
 Mid-Term Test-I (July-Dec. 2015)
 B.E. I Year MA 1056 : Mathematics- II

Max. Marks: 20

Time: 01 Hour

Note: Attempt any four questions. All questions carry equal marks.

Q.1	Solve $\frac{dy}{dx} = \frac{x^2+xy}{x^2+y^2}$	(05)
Q.2(i)	Solve $6y^2dx - x(2x^3 + y)dy = 0$	2.5+2.5
(ii)	Find the inverse of $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$	
Q.3	Find non-singular matrices P and Q such that PAQ is in the normal form of A where $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$	(05)
Q.4	Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$	(05)
Q.5	Show that three equations $-2x + y + z = a$ $x - 2y + z = b$ $x + y - 2z = c$ have no solution unless $a + b + c = 0$ in which case they have infinitely solutions find these solutions when $a = 1, b = 1, c = -2$	(05)

Department of Applied Mathematics & Computational Science

B.E. I Year

Mid-Term Test-I(Dec.2013-April 2014)

Mathematics II : MA 1056

[Max. Marks: 20]

Note: Attempt any FOUR questions. All questions carry equal marks :

Q.1(i)	Find the inverse of A where $\begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$	(03)
(ii)	Solve the following differential equation: $\frac{dy}{dx} = \frac{y+x-2}{y-x-4}$	(02)
Q.2	Reduce the Matrix A to its Normal form, where $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ & hence find the Rank of the Matrix.	(05)
Q.3	Solve completely the following system of equations: $\begin{aligned} 2w + 3x - y - z &= 0 \\ 4w - 6x - 2y + 2z &= 0 \\ -6w + 12x + 3y - 4z &= 0 \end{aligned}$	(05)
Q.4	Solve $\frac{dy}{dx} = \frac{y+1}{(y+2)e^x - x}$	(05)
Q.5	Solve $\{ 2xy \cos x^2 - 2xy + 1 \} dx + \{ \sin x^2 - x^2 + 3 \} dy = 0$	(05)

Mid-Term Test I (Jan.-June 2015)
B.E. I Year
Mathematics II: MA 1056

[Max. Marks: 20]

Time: 1 hour]

Note : Attempt any four questions . All questions carry equal marks.

Q.1	Solve. $(x \sec^2 y - x^2 \cos y) dy = (\tan y - 3x^4) dx$	(05)
Q.2(i)	Solve. $x \log x \frac{dy}{dx} + y = 2 \log x$.	(2.5+2.5)
(ii)	Find the nature, index and signature of the given quadratic form $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_1x_3$	(05)
Q.3	Find the rank of the following matrices by reducing it into normal form $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$	(05)
Q.4	Find the eigen values and eigen vectors of the following matrix $\begin{bmatrix} 6 & 2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	(05)
Q.5	Solve. $\frac{dy}{dx} = 2y \tan x + y^2 \tan^2 x$.	(05)

$$\frac{dy}{dx} + y f(x) = (1) \quad y e^{y/2} = y$$

Time : 1hr.]

Note : Attempt any FOUR questions.

Q.1 Reduce the Matrix A to the normal form,

$$A = \begin{bmatrix} 3 & -4 & -1 & 2 \\ 1 & 7 & 3 & 1 \\ 5 & -2 & 5 & 4 \\ 9 & -3 & 7 & 7 \end{bmatrix}$$

Hence find the rank of A

Rank - 2

(05)

Q.2

If $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 9 & 3 \\ 1 & 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, Verify that

(05)

$$(AB)^{-1} = B^{-1}A^{-1}$$

Q.3

Find the rank of matrices

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(05)

Q.4

Solve the differential equation

(05)

(i) $y' + 4xy + xy^3 = 0$

(ii) $(y^2x + 2x^2y^3)dx + (x^2y - x^3y^2)dy = 0$

Q.5

Solve. $(xy^2 - x^2)dx + (3x^2y^2 + x^2y - 2x^3 + y^2)dy = 0$

(05)

$R_1 - R_3$ $-1+2$

$$\begin{bmatrix} 2 & -3 & -5 \\ -1 & -2 & -4 \\ 1 & 3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & -3 & -5 \\ 1 & -1 & -2 & -4 \\ 0 & 1 & 1 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$R_{eq} \rightarrow R_1 + R_3$

$$\begin{bmatrix} 6 & 3 & 0 & -7 \\ 1 & -1 & -2 & -4 \\ 6 & 3 & 0 & -7 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 3 & 0 & -7 \\ 1 & -1 & -2 & -4 \\ 6 & 3 & 0 & -7 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 3 & 0 & -7 \\ 1 & -1 & -2 & -4 \\ 6 & 3 & 0 & -7 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

~~$R_1 + R_3 + R_2$~~

~~$(R_1 + R_3) - R_2$~~

~~$4 \ 3 \ 2 \ 3$~~

~~$5 \ -2 \ 5$~~

Class Test -I (Mar.- 2013)

1AM051- Applied Mathematics -II

For (IT.& Civil)

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Time :70 min.

Maximum Marks:25

Note : Attempt all questions.

Q1

Find the inverse of $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ by Partition method. *eigen value & eigen vector*

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Q2. For what values of k the equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ has a solution.

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Q3. Find the Fourier series for the Function $f(x) = \left(\frac{\pi - x}{2}\right)^2$ in $(0, 2\pi)$.

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Q4. Expand the function $f(x) = x \sin x$ as a Fourier series in the interval $-\pi \leq x \leq \pi$.

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